Fisher Information Loss for Coordination Rules in Decentralized Sensor-Agent Networks

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Abstract

Decentralized sensor networks routinely face a last-mile coordination problem: each mote or edge agent observes only a local reading and a local coordinate, yet the team must decide *who exchanges what with whom*, and when to *quarantine* a report under uncertainty. Perception—local categorical distributions induced by readings and geometry—and coordination—a deterministic rule that compresses the joint into exchange or quarantine actions—are often designed separately. While Dec-POMDP active perception and information-driven sensing carefully treat what to observe, the coordination layer is frequently heuristic and evaluated only by task reward or throughput. This paper asks a complementary question: *holding the agents' local laws fixed, which θ-independent coordination rules destroy the least Fisher information about the global field or event parameters?*

We formalize coordination as an AggregationMap $A = f(Z)$ and quantify information loss by the data-processing gap $I_Z(\theta) - I_A(\theta) \succeq 0$ between the Fisher information of a declared product joint and that of its pushforward. Rules are ranked by $\mathrm{tr}(\Delta) = \mathrm{tr}(I_Z - I_A)$. The intended primary application is decentralized sensor networks. On a synthetic unit-disk 2D field (9 motes) the core order holds and quarantine-wider is the lowest-loss non-trivial rule; on a 12-mote Intel Berkeley Lab (IBRL) real-data control the core order also holds, but both quarantine rules lose more than baseline, so the quarantine safety valve pays an information cost and the full ranking is data-dependent. As a reproducible analytic illustration of the same AggregationMap programme, we also use the Jevometry three-agent data-cleaning council (applicability, repair, risk) and compare seven pure rules—baseline, stricter, wider, quarantine-to-review, a hybrid quarantine-wider split, coarse binary, and a near-identity upper bound—on a grid over evidence and ambiguity. At the design centre, baseline coordination discards roughly half of $\mathrm{tr}(I_Z)$; softening quarantine into human review recovers a substantial fraction of that loss, and combining that softening with distinct auto-normalize/auto-impute labels yields a further improvement that holds on the full design grid. The ordering near-identity $\leq$ wider $\leq$ baseline $\leq$ stricter is stable on every grid cell. We discuss design implications for sensor-agent exchange and quarantine policies, API constraints that force deterministic maps, and the role of the cleaning numbers as an analytic illustration alongside the unit-disk and IBRL results. Accompanying code and reproducible tables ship with the Jevometry repository.

1 Introduction

Modern sensing stacks make it cheap for each mote or edge agent to emit a calibrated-looking categorical distribution over local judgments: observation quality, whether to report, which neighbour to exchange with, or whether a reading should be quarantined. In a decentralized sensor network these local judgments are rarely the end product. A coordinator—often a short piece of deterministic logic shared by the team—compresses the joint of agent outputs into exchange decisions (who/what to exchange) or a quarantine safety valve [Jevometry maintainers, 2026].

This compression is an information bottleneck. Classical decision theory and Dec-POMDP planning optimize rewards defined on beliefs or on prediction quality [Lauri and Oliehoek, 2020, Lauri et al., 2020, Oliehoek and Amato, 2016]. Information-theoretic multi-agent sensing maximizes mutual information or related set functions, often with submodular greedy guarantees [Krause et al., 2008, Nemhauser et al., 1978, Corah and Michael, 2021, Shi et al., 2026]. Those lines of work choose *what to observe* or *who should observe*. They do not, by themselves, tell a practitioner which of two already-deployed coordination rules retains more sensitivity to the global field or event parameters θ the network was designed to track.

Fisher information provides a local, differentiable language for that sensitivity [Fisher, 1925, Amari, 2016]. For a discrete model $P_\theta(Z)$ with score Jacobian J_Z, the Fisher matrix $I_Z = J_Z^\top \mathrm{diag}(P_\theta)^{-1} J_Z$ measures how strongly infinitesimal parameter moves change the outcome distribution. When a fixed map f produces actions $A = f(Z)$, the data-processing inequality for Fisher information asserts $I_Z - I_A \succeq 0$ under standard regularity conditions [Zamir, 1998, Cover and Thomas, 2006]. The scalar $\mathrm{tr}(I_Z - I_A)$ is then a comparable loss summary when I_Z is held fixed.

Target setting and current evidence. Our primary narrative is decentralized sensor agents under local observations, using Fisher ΔI as a shared decision function to choose among coordination rules—including quarantine when uncertain. The intended main experiment is a synthetic unit-disk field in 2D; a secondary control is a 12-mote Intel Berkeley Lab (IBRL) subnet. Both are reported in Section 6.5. We additionally report a reproducible analytic instantiation of the same AggregationMap programme: the Jevometry three-agent cleaning council, whose tables and figures serve as a proof of concept for ranking `near`/`wider`/`baseline`/`stricter`/`quarantine_*`/`near_identity`-style rules by $\mathrm{tr}(\Delta)$.

Contributions.

1. We cast multi-agent coordination-rule comparison as ranking θ-independent aggregation maps by Fisher information loss under a shared declared joint, with sensor-network exchange and quarantine as the motivating AggregationMap semantics (Section 3).

2. We map the programme onto three layers—global field parameters θ, local agent categorical laws, and θ-independent AggregationMaps—and run a synthetic unit-disk main experiment and an IBRL real-data control (Sections 4 and 6.5). The core order holds on both; quarantine-wider is best on unit-disk (1.784), whereas on IBRL both quarantine rules lose more than baseline (3.042, 3.101 vs. 2.785).

3. As a current analytic instantiation, we instantiate seven pure rules in the Jevometry cleaning council and provide a reproducible harness that sweeps a parameter grid, exports CSV/JSON (including parameter-axis loss diagonals and fractional losses), and emits publication figures (Section 5).

4. Empirically, at the design centre the baseline rule loses about 44% of $\mathrm{tr}(I_Z)$; quarantine-to-review cuts loss by roughly a quarter relative to baseline; a hybrid quarantine-wider rule improves further ($\mathrm{tr}(\Delta) \approx 2.16$) and dominates quarantine-to-review on the full grid; a binary gate loses more than stricter mid-risk rejection (Section 6). These numbers illustrate the AggregationMap audit analytically; sensor results are in Section 6.5.

5. We discuss how this metric complements reward-centric Dec-POMDP evaluation for sensor coordination and what API constraints (deterministic maps only) imply for future learned coordinators (Section 7).

This manuscript is intended as a short journal / arXiv draft accompanying open-source experiments. Cleaning-council numbers are analytic and illustrative; the unit-disk and IBRL sensor results are the primary empirical story.

2 Related work

Dec-POMDP active perception. Lauri and Oliehoek [2020] reduce multi-agent active perception with prediction rewards to standard Dec-POMDP solvers via individual prediction actions, with decentralization-loss bounds. Lauri et al. [2020] optimize convex belief-dependent rewards (e.g., negative entropy) with policy graph improvement in discrete and continuous-state settings. Both lines typically treat the team as fixed and fold information objectives into rewards rather than post-hoc comparing deterministic coordinators.

Information-driven multi-agent sensing. Submodular sensor selection and multi-robot tracking maximize mutual information or related coverage objectives with greedy $(1 - 1/e)$ guarantees under monotonicity [Krause et al., 2008, Corah and Michael, 2021]. Olkin et al. [2024] and Molloy and Nair [2023] further connect planning to information and smoother entropy.

IMAS^2 (closest Dec-POMDP relative). Shi et al. [2026] study *joint* sensing-agent selection and decentralized active-perception policy synthesis in Dec-POMDPs (`arXiv:2510.20009`; AAMAS 2026). An inner layer maximizes mutual information between latent trajectories (or a secret property) and the team's partial observations; an outer layer greedily selects agents under submodularity, with a $(1 - 1/e)$-style guarantee when successive marginal gains satisfy their stated ratio condition. Experiments use multi-agent cooperative perception in a grid world. That programme *optimizes who senses and how* under Dec-POMDP dynamics. Ours is a related but distinct route: we *hold* the local agent laws fixed and rank θ-independent AggregationMaps by Fisher information loss $\Delta_f = I_Z - I_A$ after action aggregation—including quarantine as a deterministic safety valve—rather than maximizing mutual information via submodular agent selection. The two metrics answer complementary questions (select/synthesize sensors vs. audit last-mile coordination).

Information geometry for agents. Fisher–Rao geometry supplies local metrics on parametric families of distributions [Amari, 2016]. Jevometry operationalizes node- and system-level Fisher diagnostics for agent systems [Jevometry maintainers, 2026]. We use its pushforward `information_loss` primitive and do not modify the underlying DECLARED_PRODUCT construction.

3 Problem formulation

3.1 Declared product joint

Let $\theta \in \Theta \subset \mathbb{R}^d$ parameterize a multi-agent system. Each agent $i \in \{1, \dots, m\}$ emits a categorical distribution $P_\theta(Z_i)$ over a finite local alphabet $\mathcal{Z}_i$. A *declared product joint* asserts conditional independence given θ:

$$P_\theta(Z = z) = \prod_{i=1}^{m} P_\theta(Z_i = z_i), \qquad z = (z_1, \dots, z_m) \in \mathcal{Z} = \prod_i \mathcal{Z}_i. \tag{1}$$

In the cleaning instantiation below, $m = 3$, $d = 2$, and $\theta = (\text{evidence}, \text{ambiguity})$ with coordinates in $[0, 1]$. In the planned sensor instantiation, θ instead denotes global field or event parameters (e.g., source location, intensity, or event indicators), not raw single-node readings. The product

declaration is an assumption of the model, not an empirical claim about live APIs or radio traces.

3.2 Coordination as aggregation

Definition 1 (Coordination rule)**.** *A coordination rule is a map $f : \mathcal{Z} \to \mathcal{A}$ into a finite action alphabet $\mathcal{A}$. We call f* pure *if it is a total function of the joint outcome alone (no hidden randomness) and θ-*independent *if f does not depend on θ.*

The induced action law is the pushforward

$$P_\theta(A = a) = \sum_{z : f(z) = a} P_\theta(Z = z). \tag{2}$$

If $J_Z(\theta)$ is the Jacobian of the joint probability vector with respect to θ, the aggregate Jacobian J_A is obtained by summing rows of J_Z that share an action label.

3.3 Fisher information loss

Definition 2 (Information loss of a rule)**.** *For a θ-independent rule f, define*

$$I_Z(\theta) = J_Z(\theta)^\top \operatorname{diag}\big(P_\theta(Z)\big)^{-1} J_Z(\theta), \tag{3}$$

$$I_A(\theta) = J_A(\theta)^\top \operatorname{diag}\big(P_\theta(A)\big)^{-1} J_A(\theta), \tag{4}$$

$$\Delta_f(\theta) = I_Z(\theta) - I_A(\theta). \tag{5}$$

We report $\operatorname{tr}(I_Z)$, $\operatorname{tr}(I_A)$, and $\operatorname{tr}(\Delta_f)$ as scalar summaries, and we verify $\Delta_f \succeq 0$ numerically (PSD check) as a data-processing diagnostic.

Proposition 1 (DPI for fixed maps)**.** *Under standard positivity and differentiability conditions, any θ-independent f satisfies $\Delta_f(\theta) \succeq 0$. Equality holds when f is injective on the support of P_θ.*

Remark 1. *Maps that depend on θ (soft policies, learned routers conditioned on parameters) fall outside this guarantee in the Jevometry implementation and are reported as unsupported. This paper therefore restricts attention to pure, fully covering, θ-independent rules—the class that matches typical production if-then coordinators, mote firmware exchange policies, and the library's AggregationMap contract.*

Comparison principle. Fix the agents and the joint construction. Rank rules by $\operatorname{tr}(\Delta_f)$ (lower is better) or equivalently by $\operatorname{tr}(I_A)$ (higher is better). Because I_Z is shared, ranking by loss is identical to ranking by retained information.

4 Three layers for sensor-agent coordination

The AggregationMap programme admits a three-layer reading that matches decentralized sensing:

(i) **Global parameters θ.** θ denotes field- or event-level quantities of interest (source location, intensity, spatial covariance hyperparameters, binary event indicators), *not* the raw scalar reading at a single mote. Fisher loss is measured with respect to this θ.

(ii) **Local agent laws.** Each sensor is an agent. Its categorical distribution $P_\theta(Z_i)$ is induced by local coordinates and local readings—for example, bins of observation quality, a report/hold decision, or a neighbour-choice alphabet under a unit-disk communication graph. The declared product joint assembles these local laws.

(iii) **AggregationMap as exchange and quarantine.** A θ-independent rule f compresses the joint into coordination actions: who exchanges with whom, what payload to send, or whether to quarantine a report when local uncertainty is high. Candidate families reuse the same qualitative knobs studied below—near-neighbour exchange, wider neighbourhoods, baseline firmware, stricter thresholds, quarantine variants, and a near-identity upper bound—and are ranked by $\mathrm{tr}(\Delta_f) = \mathrm{tr}(I_Z - I_A)$.

Sensor experiments.

- **Main:** synthetic unit-disk sensors in a 2D field (9 motes, 3×3 grid, $R = 0.45$). Local alphabets and AggregationMaps follow layers (ii)–(iii); results are in Table 2.
- **Control:** Intel Berkeley Lab (IBRL) subnet of 12 motes (ids 4, 11, 16, 18, 20, 26, 34, 39, 42, 47, 50, 53), as a real-deployment sanity check under the same AggregationMap interface; results are in Table 3.

The cleaning-council study in Sections 5–6 is the present, fully reproducible analytic illustration of layers (i)–(iii) with $\theta = (\text{evidence}, \text{ambiguity})$ and repair/quarantine actions standing in for exchange/quarantine semantics.

5 Method and experimental design

5.1 Three-agent cleaning council (analytic instantiation)

We reuse the offline Jevometry tutorial `examples/three_agent_cleaning` [Jevometry maintainers, 2026] as a proof-of-concept AggregationMap audit—not as a claim about live sensor fields. Three analytic softmax agents produce distributions for a single candidate repair of a noisy numeric field:

Agent	Primitive	Support
applicability	Noul	$\{\text{false}, \text{true}\}$
repair	Choice	$\{\text{normalize}, \text{impute}, \text{quarantine}\}$
risk	Score	$\{0, 1, 2\}$ (low/med/high)

Local logits are affine in $(\theta - \theta_0)$ around the design centre $\theta_0 = (0.7, 0.3)$, and probabilities are obtained by softmax. The joint is the declared product (ConstructionMode.DECLARED_PRODUCT). No live API is queried; no actual database write is performed. Quarantine in this council plays the same structural role as a sensor quarantine valve: a coarse action that merges distinguishable joint outcomes and therefore inflates $\mathrm{tr}(\Delta_f)$ unless softened (e.g., into human review or delayed exchange).

5.2 Coordination rules

We compare seven rules, all pure and θ-independent:

baseline Reject if inapplicable, quarantine, or high risk; auto-fix if normalize and low risk; otherwise human review.

stricter As baseline, but medium risk also rejects.

wider As baseline rejects; low-risk normalize and impute become distinct auto labels (`auto_normalize`, `auto_impute`).

quarantine_review Quarantine routes to human review instead of reject; reject only on inapplicable or high risk.

quarantine_wider Quarantine-to-review combined with wider's dual auto labels: the empirically strongest coarse policy in this study.

coarse_binary Reject under baseline reject conditions; otherwise a single `proceed` action (auto-apply and review collapsed).

near_identity One action label per joint outcome tuple (18 labels): Fisher retention upper bound under this alphabet.

In sensor language, baseline/stricter/wider correspond to narrower or broader exchange policies and risk thresholds; quarantine variants are safety valves; near-identity is an information-retention upper bound rather than a deployable radio schedule.

5.3 Harness

The script `paper_experiments.py` builds nodes and joint once, evaluates I_Z, I_A, and Δ_f on an 8×8 grid over evidence $\in [0.2, 0.9]$ and ambiguity $\in [0.1, 0.8]$, and additionally records the exact centre θ_0. Outputs:

- `papers/coord-fisher/data/rule_loss_grid.{csv,json}`
- `papers/coord-fisher/data/center_summary.json`
- `papers/coord-fisher/data/center_loss_anisotropy.csv`, `rule_loss_anisotropy_grid.csv`, fractional-loss heatmaps
- `papers/coord-fisher/data/iteration_summary.json`
- Figures 1–4 under `papers/coord-fisher/figures/`

Library code under `src/jevometry` is unchanged; only example-level rules and scripts are extended. The unit-disk and IBRL harnesses reuse the same AggregationMap and `information_loss` interface.

6 Experiments

6.1 Centre-point ranking

Table 1 reports traces at θ_0. All rules return status `ok` with PSD loss matrices. The near-identity map loses nothing (up to numerical noise), confirming the implementation against the equality case of the DPI. Baseline loses $\mathrm{tr}(\Delta) \approx 3.03$ out of $\mathrm{tr}(I_Z) \approx 6.84$—about 44%. Wider slightly improves on baseline by preserving distinct auto paths. Stricter mid-risk rejection increases loss. Softening quarantine into review yields a large practical gain ($\mathrm{tr}(\Delta) \approx 2.27$); stacking that softening with wider's dual auto labels (`quarantine_wider`) improves further to ≈ 2.16 and is the best non-identity coarse policy here. Collapsing to a binary gate is worst among the listed rules (≈ 3.71).

The core ordering highlighted in prior exploratory work remains:

$$\text{near-identity} \;\le\; \text{wider} \;\le\; \text{baseline} \;\le\; \text{stricter}$$

on *every* cell of the 8×8 design grid (64/64), not only at the centre (numerical tolerances 10^{-9}). Quarantine-to-review inserts *below* wider on the loss axis; `quarantine_wider` inserts still lower and satisfies $\mathrm{tr}(\Delta_{\text{quarantine_wider}}) \le \mathrm{tr}(\Delta_{\text{quarantine_review}})$ on 65/65 recorded centre-and-grid evaluations. Coarse binary sits *above* stricter—an informative deviation from the original four-rule story.

Table 1: Fisher traces at centre $\theta_0 = (0.7, 0.3)$, sorted by ascending loss. $|A|$ is the number of actions with positive mass under the pushforward. Cleaning-council analytic illustration—not unit-disk or IBRL.

Rule	$\mathrm{tr}(I_Z)$	$\mathrm{tr}(I_A)$	$\mathrm{tr}(\Delta)$	$\lvert A\rvert$	Status
near_identity	6.8418	6.8418	0.0000	18	ok
quarantine_wider	6.8418	4.6816	2.1602	4	ok
quarantine_review	6.8418	4.5764	2.2654	3	ok
wider	6.8418	3.8285	3.0133	4	ok
baseline	6.8418	3.8080	3.0338	3	ok
stricter	6.8418	3.4198	3.4220	3	ok
coarse_binary	6.8418	3.1296	3.7122	2	ok

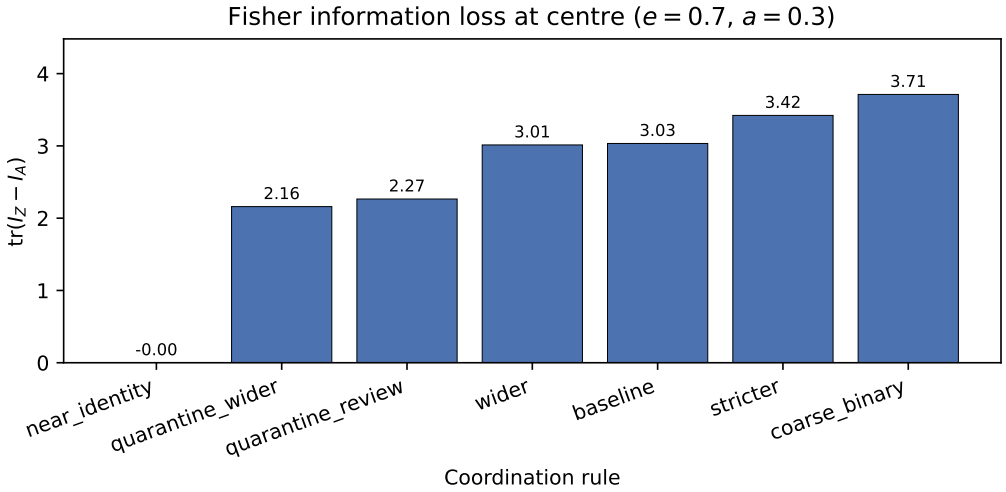


Figure 1: Information loss $\mathrm{tr}(I_Z - I_A)$ by rule at the design centre (cleaning-council illustration).

6.2 Grid behaviour

Figure 2 shows loss heatmaps for five non-identity rules over the parameter grid. Absolute loss varies with θ because both $I_Z(\theta)$ and the mass allocation across f-fibres change, but the qualitative ranking is stable in large regions of the domain: softer, finer action alphabets sit cooler (lower loss) than stricter or binary gates. Figure 3 slices loss against evidence at ambiguity nearest 0.3; near-identity stays at zero, while coarse binary and stricter remain the most lossy.

6.3 Action masses

Figure 4 shows how probability mass is partitioned at the centre. Stricter reallocates medium-risk mass into reject. Wider splits auto-apply into normalize versus impute. Quarantine-to-review shrinks reject in favour of human review. Coarse binary folds auto-apply and review into `proceed`. These mass shifts are the mechanism behind Fisher loss: merging outcomes with dissimilar score rows collapses distinguishable parameter directions—the same mechanism AggregationMaps will exhibit when merging exchange or quarantine fibres in a sensor field.

6.4 Anisotropy and the quarantine-wider mechanism

Trace summaries hide which parameter direction is discarded. At θ_0, the diagonal entries of Δ_f in the $(\text{evidence}, \text{ambiguity})$ basis show that `quarantine_review` cuts ambiguity-axis loss roughly

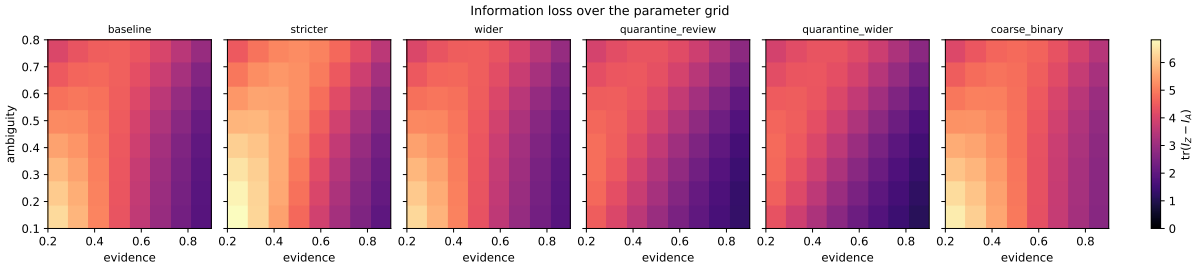


Figure 2: Loss heatmaps over evidence×ambiguity for five coordination rules (cleaning-council illustration).

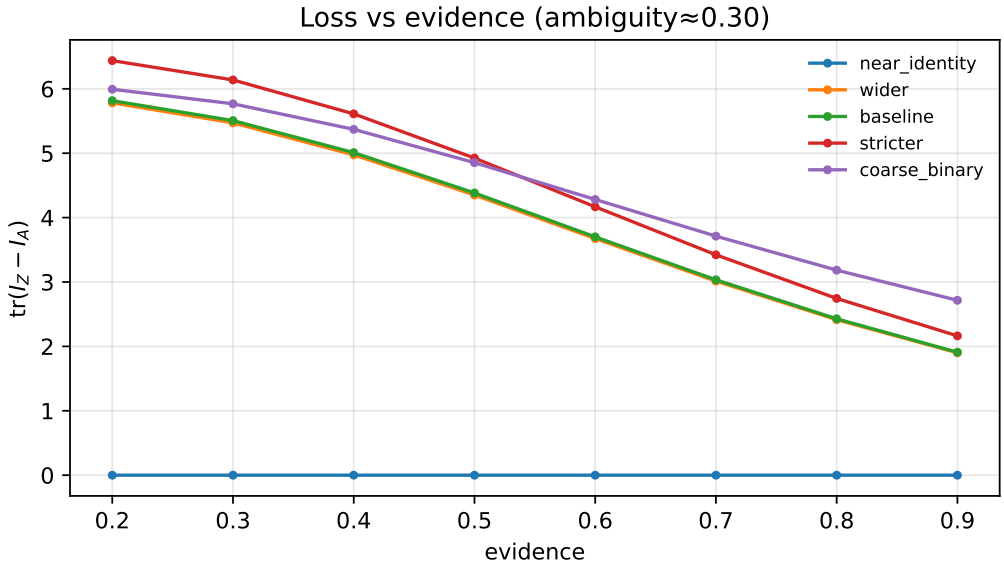


Figure 3: Loss versus evidence at fixed ambiguity near 0.3 (cleaning-council illustration).

in half relative to baseline ($\Delta_{aa} \approx 0.95$ vs ≈ 1.69) while evidence-axis loss stays comparable. Relative losses $\mathrm{tr}(\Delta)/\mathrm{tr}(I_Z)$ preserve the same ranking where $I_Z(\theta)$ varies across the grid.

Pushforward masses explain the hybrid gain: relative to `quarantine_review`, `quarantine_wider` replaces a single `auto_apply` atom by distinct `auto_normalize` and `auto_impute` atoms and moves additional mass out of `human_review` into `auto_impute`; `reject` mass is unchanged. Finer auto fibres retain score rows that the coarser review rule had merged.

6.5 Sensor-network results

Unit-disk main experiment. We place $N = 9$ motes on a 3×3 grid in $[0, 1]^2$ with unit-disk radius $R = 0.45$ (12 edges, mean degree 2.67). The global field parameter is $\theta = (\text{amplitude}, \text{length scale}) = (0.7, 0.35)$ of a Gaussian bump with fixed centre. The same seven AggregationMaps are applied with sensor exchange/quarantine semantics. The Fisher approximation check passes and the core order near-identity $\leq$ wider $\leq$ baseline $\leq$ stricter holds. Among non-trivial rules, `quarantine_wider` has the lowest loss (1.784), ahead of `wider` (1.820) and `baseline` (1.833); `coarse_binary` (2.246) and `stricter` (2.833) lose most (Table 2).

IBRL real-data control. We take a 480-row subsample of the Intel Berkeley Lab labdata[1] over 12 motes (ids 4, 11, 16, 18, 20, 26, 34, 39, 42, 47, 50, 53) selected by farthest-point

[1] https://db.csail.mit.edu/labdata/labdata.html

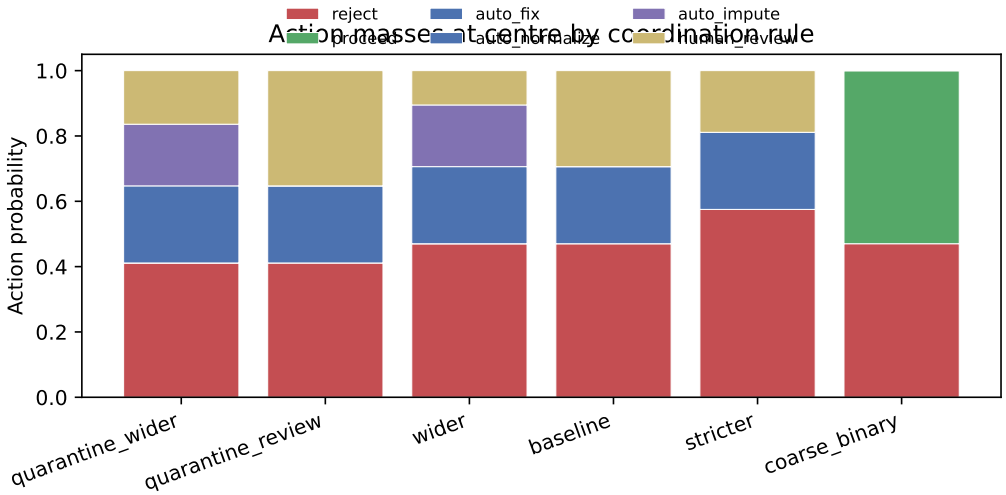


Figure 4: Pushforward action probabilities at the centre for non-identity rules (cleaning-council illustration).

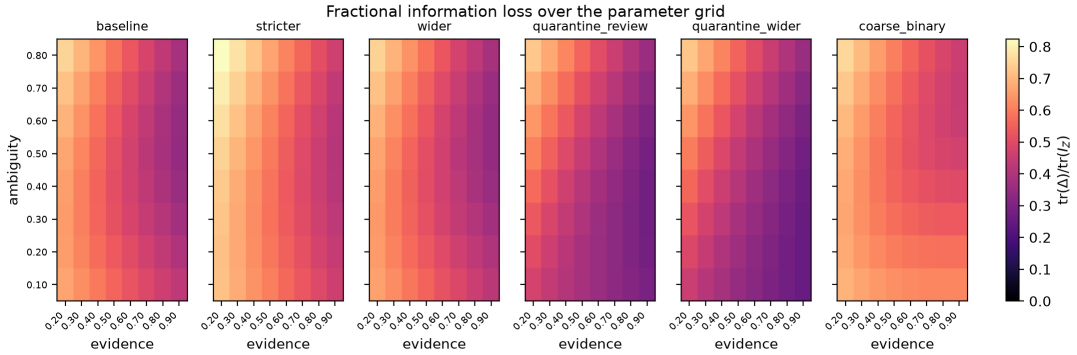


Figure 5: Fractional loss $\mathrm{tr}(\Delta)/\mathrm{tr}(I_Z)$ over the evidence×ambiguity grid for non-identity rules (cleaning-council illustration).

sampling, with θ the MinMax-normalized mean temperature and humidity and a unit-disk graph at normalized radius 0.35 (6 edges). The Fisher approximation check passes and the core order again holds, but the full ordering differs from unit-disk (Table 3): `near_identity` (0.000) < `wider` (2.784) ≈ `baseline` (2.785) ≈ `coarse_binary` (2.796) < `quarantine_wider` (3.042) < `quarantine_review` (3.101) < `stricter` (3.474).

Reading the two tables honestly. On IBRL both quarantine rules lose more than baseline (3.042 and 3.101 versus 2.785)—the reverse of unit-disk, where `quarantine_wider` is the best non-trivial rule. The quarantine safety valve therefore pays an information cost on real data; whether it helps or hurts $\mathrm{tr}(\Delta)$ is data-dependent, and we do not claim a universal ranking beyond the core order. On IBRL, `wider`, `baseline`, and `coarse_binary` are nearly tied (2.78–2.80), so these three rules are practically indistinguishable by this metric on that subnet.

7 Discussion

Sensor-network reading. Under layers (i)–(iii), $\mathrm{tr}(\Delta_f)$ is a shared offline score for choosing among exchange and quarantine rules when local agent laws are held fixed. Softer quarantine (delay or human/operator review instead of hard drop) and finer exchange alphabets (distinct

Table 2: Synthetic unit-disk 2D field ($N = 9$, $R = 0.45$), centre-point traces at $\theta = (0.7, 0.35)$.

Rule	$\mathrm{tr}(I_Z)$	$\mathrm{tr}(I_A)$	$\mathrm{tr}(\Delta)$
`near_identity`	5.788	5.788	0.000
`wider`	5.788	3.969	1.820
`baseline`	5.788	3.955	1.833
`stricter`	5.788	2.955	2.833
`quarantine_review`	5.788	3.913	1.875
`quarantine_wider`	5.788	4.004	1.784
`coarse_binary`	5.788	3.542	2.246

Table 3: IBRL control (12 motes, 480 rows), centre-point traces at the feature means.

Rule	$\mathrm{tr}(I_Z)$	$\mathrm{tr}(I_A)$	$\mathrm{tr}(\Delta)$
`near_identity`	3.943	3.943	0.000
`wider`	3.943	1.159	2.784
`baseline`	3.943	1.158	2.785
`stricter`	3.943	0.469	3.474
`quarantine_review`	3.943	0.842	3.101
`quarantine_wider`	3.943	0.901	3.042
`coarse_binary`	3.943	1.147	2.796

neighbour or payload labels) are the direct analogues of the quarantine-wider gains in Table 1. Binary proceed/reject gates remain operationally convenient but informationally costly. The sensor results temper this: quarantine-wider is best on unit-disk but both quarantine rules lose more than baseline on IBRL (Section 6.5). The ranking machinery and DPI certificate transfer across settings; the absolute traces and the position of quarantine in the ranking do not.

What "half the Fisher" means. A 44% drop in $\mathrm{tr}(I_Z)$ after baseline coordination does not mean the system is 44% "wrong." It means that, locally, infinitesimal changes in the parameters move the final action distribution less than they move the joint of agent outputs. If a monitoring dashboard only watches action or exchange rates, it understates how sensitive the underlying agents are. If a downstream estimator tries to invert actions for θ, it faces a strictly harder problem than inverting Z.

Design takeaway. Among implementable three/four-action policies in the cleaning illustration, routing quarantine to review retained more Fisher than baseline or wider alone; combining that softening with distinct auto-normalize/auto-impute labels is strictly better still on this grid. Practitioners can treat $\mathrm{tr}(\Delta_f)$—and, when needed, its parameter-axis diagonals—as a cheap offline score when debating rule changes before A/B tests or firmware flashes.

Relation to Dec-POMDP rewards and IMAS^2. Prediction rewards and negative-entropy objectives [Lauri and Oliehoek, 2020, Lauri et al., 2020] optimize perception policies. IMAS^2 [Shi et al., 2026] further couples agent selection with mutual-information-maximizing sensing policies under Dec-POMDP dynamics and submodular greedy selection. Our metric does not replace those programmes; it audits the *last* deterministic compression after perception (and any selection) is fixed. Selection expands who may observe; Fisher Δ_f ranking asks how much of that observation survives a chosen AggregationMap—including quarantine—at the exchange layer.

API constraints. Many agent and mote platforms expose only sampled outcomes plus deterministic business or firmware rules. Soft, θ-dependent routers would require either a declared conditional action model or a refusal of the DPI certificate—exactly the contract encoded by Jevometry's AggregationMap. Extending the present study to stochastic coordinators is future work and needs additional regularity.

8 Limitations

- The reported empirical numbers come from hand-specified analytic softmax families in the cleaning council, not from fitted live decision models. Absolute traces are illustrative.
- The sensor experiments are small: 9 synthetic motes and a 12-mote, 480-row IBRL subsample at a single centre point. The quarantine ranking flips between them, so rankings beyond the core order are data-dependent and should not be read as universal.
- Conditional independence is declared, not tested.
- We report parameter-axis diagonals and eigenvalues of Δ_f, but full CRLB-style reads would still need an observation likelihood contract the cleaning example does not provide.
- Action alphabets differ across rules; $\mathrm{tr}(I_A)$ comparisons remain meaningful as retained information about θ, but interpretations that treat $|A|$ as utility are out of scope.
- No human, task-reward, or sensing-accuracy study yet validates that lower Fisher loss improves cleaning or field-estimation outcomes.

9 Conclusion

We proposed ranking multi-agent coordination rules by Fisher information loss under a shared declared joint, with decentralized sensor-agent exchange and quarantine as the primary AggregationMap story. Holding local perception fixed, deterministic coordinators can discard a large fraction of the joint Fisher trace; small rule edits (quarantine handling, finer action alphabets, risk or neighbourhood thresholds) move that loss by amounts that are easy to measure offline. The three-agent cleaning council provides a reproducible analytic proof of concept—including the stable ordering near-identity $\leq$ wider $\leq$ baseline $\leq$ stricter and the quarantine-wider gains. On sensor networks the core order holds on both a synthetic unit-disk field and a 12-mote IBRL control; quarantine-wider is best on unit-disk, but on IBRL both quarantine rules lose more than baseline, so the quarantine safety valve pays an information cost on real data and the full ranking is data-dependent.

Future work includes (i) scaling the unit-disk and IBRL experiments to more motes, centre points, and time windows, (ii) replacing analytic agents with TypeSafe / live decision-model Jacobians where appropriate, (iii) stochastic coordinators with explicit conditional action models, and (iv) coupling $\mathrm{tr}(\Delta_f)$ with task-level utilities (field estimation error, cleaning quality) in JAAMAS-style empirical studies.

Reproducibility

From the repository root:

```
uv sync --locked
uv pip install matplotlib   # paper figures
uv run python examples/three_agent_cleaning/compare_rules.py
uv run python examples/three_agent_cleaning/paper_experiments.py
```

Artefacts land in `papers/coord-fisher/data` and `figures/`. Cleaning-council tables and figures in this PDF are the current analytic illustration; unit-disk and IBRL artefacts are `data/unit_disk_*` and `data/ibrl_*`. This PDF was built with pdflatex from `papers/coord-fisher/main.tex`.

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